

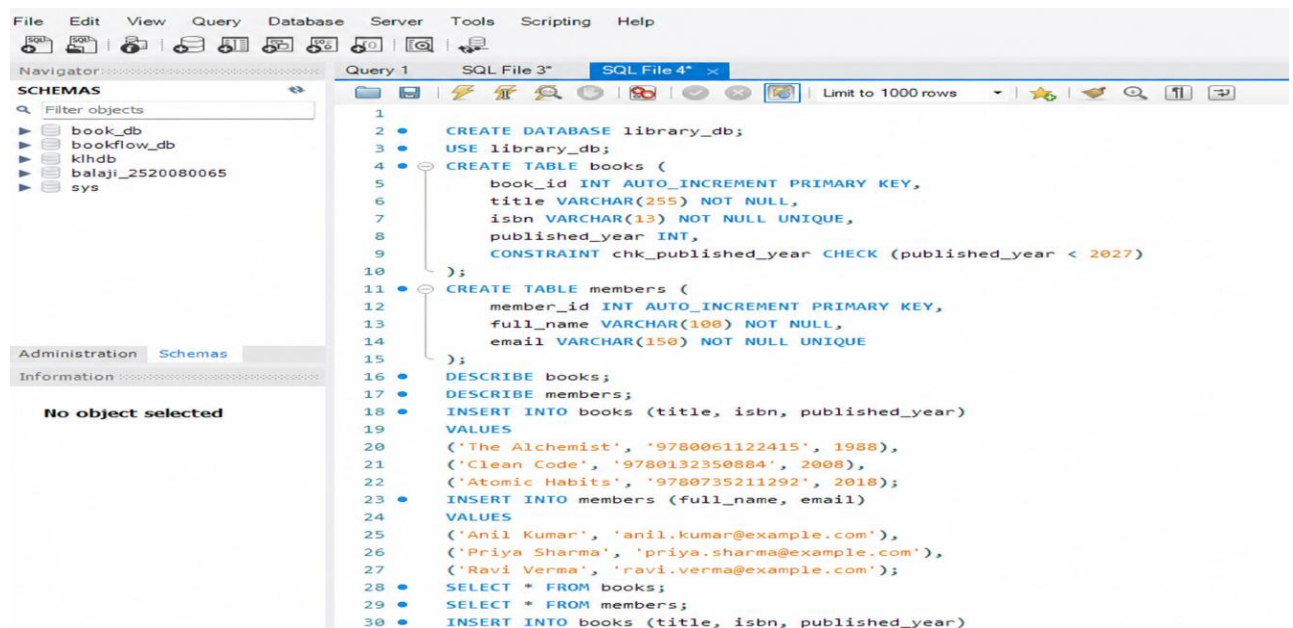
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Course: DBMS Practical week 1&2

Practical

Week 1:

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named bookflow_db. Then, design two tables: a Books table containing the columns book_id, title, isbn, and published_year, and a Members table containing the columns member_id, full_name, and email. Ensure data integrity by setting book_id and member_id as PRIMARY KEYS, applying a NOT NULL constraint to the title column so that every book must have a name, enforcing UNIQUE constraints on both isbn and email to prevent duplicate entries, and adding a CHECK constraint to ensure that the published_year is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the Books and Members tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as CREATE DATABASE, CREATE TABLE, and INSERT statements.



The screenshot shows a SQL IDE interface with a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar contains a 'SCHEMAS' panel with a search filter and a list of databases: book_db, bookflow_db, klhdb, balaji_2520080065, and sys. Below this is an 'Administration' tab with a 'Schemas' sub-tab, showing 'No object selected'. The main area displays a script for 'Query 1' in 'SQL File 3*'. The script includes the following SQL commands:

```
1 CREATE DATABASE library_db;
2 USE library_db;
3
4 CREATE TABLE books (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6     title VARCHAR(255) NOT NULL,
7     isbn VARCHAR(13) NOT NULL UNIQUE,
8     published_year INT,
9     CONSTRAINT chk_published_year CHECK (published_year < 2027)
10 );
11
12 CREATE TABLE members (
13     member_id INT AUTO_INCREMENT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(150) NOT NULL UNIQUE
16 );
17
18 DESCRIBE books;
19 DESCRIBE members;
20
21 INSERT INTO books (title, isbn, published_year)
22 VALUES
23 ('The Alchemist', '9780061122415', 1988),
24 ('Clean Code', '9780132350884', 2008),
25 ('Atomic Habits', '9780735211292', 2018);
26
27 INSERT INTO members (full_name, email)
28 VALUES
29 ('Anil Kumar', 'anil.kumar@example.com'),
30 ('Priya Sharma', 'priya.sharma@example.com'),
31 ('Ravi Verma', 'ravi.verma@example.com');
```

OUTPUT:

Result Grid				
	book_id	title	isbn	published_year
1		To Kill a Mockingbird	9780061120084	1960
2	1984		9780451524935	1949
3		The Hobbit	9780547928227	1937
4	NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
2	15:09:00	CREATE DATABASE library_db	1 row(s) affected	0.031 sec
3	15:09:07	USE library_db	0 row(s) affected	0.016 sec
4	15:09:07	CREATE TABLE Books (book_id INT PRIMARY KEY AUTO_INCREMENT, title VARCHAR(255) NOT NULL)	0 row(s) affected	0.031 sec
5	15:09:07	CREATE TABLE Members (member_id INT PRIMARY KEY AUTO_INCREMENT, full_name VARCHAR(255) NOT NULL)	0 row(s) affected	0.015 sec
6	15:09:07	INSERT INTO Books (title, isbn, published_year) VALUES ('To Kill a Mockingbird', '9780061120084', 1960), ('1984', '9780451524935', 1949), ('The Hobbit', '9780547928227', 1937)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:09:07	INSERT INTO Members (full_name, email) VALUES ('Asha Rao', 'asha.rao@example.com'), ('Vikram Singh', 'vikram.singh@example.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	15:11:24	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL)	Error Code: 1050. Table 'books' already exists	0.000 sec
9	15:11:39	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL)	Error Code: 1050. Table 'books' already exists	0.000 sec
10	15:12:05	CREATE DATABASE library_db	Error Code: 1007. Can't create database 'library_db'; database exists	0.000 sec

Practical week 2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

Task to be Done

- 1. The Catalog Search (JOINS):** Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
- 2. Collection Stats (Aggregations):** Write a query to find the total number of books published in each year.
Hint: Use COUNT(book_id) and GROUP BY published_year.
- 3. The Donation Transaction (ACID):** Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
- 4. Search Speed (Indexing):** Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title.

Skill Set Gained

- Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans).
- Analytical SQL: Using GROUP BY to generate business insights.
- Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

- book_db
- bookflow_db
- klhdb
- balaji_2520080065
- sys

Administration Schemas Information

No object selected

Query 1 SQL File 3* SQL File 4* SQL File 5*

```

1 CREATE DATABASE skytrack_db;
2 USE skytrack_db;
3 -- Flights table
4 CREATE TABLE Flights (
5     flight_id INT PRIMARY KEY,
6     flight_number VARCHAR(10) NOT NULL UNIQUE,
7     source VARCHAR(50) NOT NULL,
8     destination VARCHAR(50) NOT NULL,
9     departure_date DATE NOT NULL,
10    ticket_price DECIMAL(10,2) CHECK (ticket_price > 0)
11 );
12 -- Passengers table
13 CREATE TABLE Passengers (
14     passenger_id INT PRIMARY KEY,
15     passenger_name VARCHAR(100) NOT NULL,
16     email VARCHAR(100) UNIQUE
17 );
18 -- Bookings table (junction table)
19 CREATE TABLE Bookings (
20     booking_id INT PRIMARY KEY,
21     passenger_id INT NOT NULL,
22     flight_id INT NOT NULL,
23     booking_date DATE NOT NULL,

```

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- book_db
- bookflow_db
- klhdb
- balaji_2520080065
- sys

Administration Schemas

Information

No object selected

Query 1 SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

```
25 FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
26 );
27
28 -- Flight_History table (for the transaction task)
29 CREATE TABLE Flight_History (
30     history_id INT PRIMARY KEY,
31     flight_id INT NOT NULL,
32     action_type VARCHAR(50) NOT NULL,
33     action_date DATE NOT NULL,
34     FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
35 );
36 -- Insert 10 flights
37 INSERT INTO Flights (flight_id, flight_number, source, destination, departure_date, ticket_price) VALUES
38 (1, 'AI101', 'Hyderabad', 'Delhi', '2026-09-01', 4500.00),
39 (2, 'AI102', 'Delhi', 'Mumbai', '2026-09-01', 5200.00),
40 (3, 'AI103', 'Mumbai', 'Bengaluru', '2026-09-02', 3800.00),
41 (4, 'AI104', 'Bengaluru', 'Chennai', '2026-09-03', 3300.00),
42 (5, 'AI105', 'Chennai', 'Hyderabad', '2026-09-03', 2900.00),
43 (6, 'AI105', 'Hyderabad', 'Kolkata', '2026-09-03', 5600.00),
44 (7, 'AI107', 'Kolkata', 'Delhi', '2026-09-04', 6100.00),
45 (8, 'AI108', 'Delhi', 'Bengaluru', '2026-09-04', 5900.00),
46 (9, 'AI109', 'Mumbai', 'Hyderabad', '2026-09-05', 3400.00),
47 (10, 'AI110', 'Chennai', 'Mumbai', '2026-09-05', 4100.00);
```

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- book_db
- bookflow_db
- klhdb
- balaji_2520080065
- sys

Administration Schemas

Information

No object selected

Query 1 SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

```
46 (9, 'AI109', 'Mumbai', 'Hyderabad', '2026-09-05', 3300.00),
47 (10, 'AI110', 'Chennai', 'Mumbai', '2026-09-05', 4100.00);
48 -- Insert 10 passengers
49 INSERT INTO Passengers (passenger_id, passenger_name, email) VALUES
50 (201, 'Arjun Reddy', 'arjun.reddy@example.com'),
51 (202, 'Sneha Kapoor', 'sneha.kapoor@example.com'),
52 (203, 'Rohit Verma', 'rohit.verma@example.com'),
53 (204, 'Divya Menon', 'divya.menon@example.com'),
54 (205, 'Karan Malhotra', 'karan.malhotra@example.com'),
55 (206, 'Ananya Iyer', 'ananya.iyer@example.com'),
56 (207, 'Farhan Sheikh', 'farhan.sheikh@example.com'),
57 (208, 'Neha Joshi', 'neha.joshi@example.com'),
58 (209, 'Vivek Nair', 'vivek.nair@example.com'),
59 (210, 'Ritika Sharma', 'ritika.sharma@example.com');
60 -- Insert 12 bookings (10+ records)
61 INSERT INTO Bookings (booking_id, passenger_id, flight_id, booking_date) VALUES
62 (1,201,1, '2026-08-01'), (2,202,2, '2026-08-01'), (3,203,3, '2026-08-02'),
63 (4,204,4, '2026-08-02'), (5,205,5, '2026-08-03'), (6,206,6, '2026-08-03'),
64 (7,207,7, '2026-08-04'), (8,208,8, '2026-08-04'), (9,209,9, '2026-08-05'),
65 (10,210,10, '2026-08-05'), (11,201,6, '2026-08-06'), (12,202,7, '2026-08-06');
66 -- Display all tables
67 SELECT * FROM Flights;
68 SELECT * FROM Passengers;
```

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- book_db
- bookflow_db
- klhdb
- balaji_2520080065
- sys

Administration Schemas

Information

No object selected

Query 1 SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

```

67 • SELECT * FROM Flights;
68 • SELECT * FROM Passengers;
69 • SELECT * FROM Bookings;
70 -- INNER JOIN: passenger name, flight number, source, destination
71 • SELECT p.passenger_name AS Passenger_Name, f.flight_number AS Flight_Number,
72       f.source AS Source, f.destination AS Destination
73 FROM Bookings b
74 INNER JOIN Passengers p ON b.passenger_id = p.passenger_id
75 INNER JOIN Flights f ON b.flight_id = f.flight_id;
76 -- GROUP BY: total flights per destination
77 • SELECT destination, COUNT(flight_id) AS Total_Flights
78 FROM Flights
79 GROUP BY destination
80 ORDER BY destination;
81 -- Transection: add flight + log to history atomically
82 • START TRANSACTION;
83 • INSERT INTO Flights (flight_id, flight_number, source, destination, departure_date, ticket_price)
84 VALUES (11, 'AI111', 'Hyderabad', 'Pune', '2026-09-10', 3600.00);
85 • INSERT INTO Flight_History (history_id, flight_id, action_type, action_date)
86 VALUES (1, 11, 'NEW_FLIGHT_ADDED', CURDATE());
87 • COMMIT;
88 -- Index on flight_number
89 • CREATE INDEX idx_flight_number ON Flights(flight_number);

```

OUTPUT:

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

	flight_id	flight_number	source	destination	departure_date	ticket_price
1	AI101	Hyderabad	Delhi	Mumbai	2026-09-01	4500.00
2	AI102	Delhi	Mumbai	Bengaluru	2026-09-02	3800.00
3	AI103	Mumbai	Bengaluru	Chennai	2026-09-02	3200.00
4	AI104	Bengaluru	Chennai	Hyderabad	2026-09-03	2900.00
5	AI105	Chennai	Hyderabad	Kolkata	2026-09-03	5600.00
6	AI106	Hyderabad	Kolkata	Delhi	2026-09-04	6100.00
7	AI107	Kolkata	Delhi	Bengaluru	2026-09-04	5900.00
8	AI108	Delhi	Bengaluru	Hyderabad	2026-09-05	3300.00
9	AI109	Mumbai	Hyderabad	Mumbai	2026-09-05	4100.00
10	AI110	Chennai	Mumbai			
	NULL	NULL	NULL	NULL	NULL	NULL

Flights 1 x Passengers 2 Bookings 3 Result 4 Result 5 Flights 6

Result Grid			
Filter Rows:		Edit:	
Export/Import:		Wrap Cell Content:	
passenger_id	passenger_name	email	
201	Arjun Reddy	arjun.reddy@example.com	
202	Sneha Kapoor	sneha.kapoor@example.com	
203	Rohit Verma	rohit.verma@example.com	
204	Divya Menon	divya.menon@example.com	
205	Karan Malhotra	karan.malhotra@example.com	
206	Ananya Iyer	ananya.iyer@example.com	
207	Farhan Sheikh	farhan.sheikh@example.com	
208	Neha Joshi	neha.joshi@example.com	
209	Vivek Nair	vivek.nair@example.com	
210	Ritika Sharma	ritika.sharma@example.com	
NULL	NULL	NULL	

Result Grid				
Filter Rows:		Edit:		
Export/Import:		Wrap Cell Content:		
booking_id	passenger_id	flight_id	booking_date	
1	201	1	2026-08-01	
2	202	2	2026-08-01	
3	203	3	2026-08-02	
4	204	4	2026-08-02	
5	205	5	2026-08-03	
6	206	6	2026-08-03	
7	207	7	2026-08-04	
8	208	8	2026-08-04	
9	209	9	2026-08-05	
10	210	10	2026-08-05	
11	201	6	2026-08-06	
12	202	7	2026-08-06	
NULL	NULL	NULL	NULL	

Result Grid				
Filter Rows:		Export:		
		Wrap Cell Content:		
Passenger_Name	Flight_Number	Source	Destination	
Arjun Reddy	AI101	Hyderabad	Delhi	
Arjun Reddy	AI106	Hyderabad	Kolkata	
Sneha Kapoor	AI102	Delhi	Mumbai	
Sneha Kapoor	AI107	Kolkata	Delhi	
Rohit Verma	AI103	Mumbai	Bengaluru	
Divya Menon	AI104	Bengaluru	Chennai	
Karan Malhotra	AI105	Chennai	Hyderabad	
Ananya Iyer	AI106	Hyderabad	Kolkata	
Farhan Sheikh	AI107	Kolkata	Delhi	
Neha Joshi	AI108	Delhi	Bengaluru	
Vivek Nair	AI109	Mumbai	Hyderabad	
Ritika Sharma	AI110	Chennai	Mumbai	

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	destination	Total_Flights			
▶	Bengaluru	2			
	Chennai	1			
	Delhi	2			
	Hyderabad	2			
	Kolkata	1			
	Mumbai	2			

Result Grid							Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	flight_id	flight_number	source	destination	departure_date	ticket_price				
▶	5	AI105	Chennai	Hyderabad	2026-09-03	2900.00				
*	NULL	NULL	NULL	NULL	NULL	NULL				